TESTIMONY OF JARED RODRIGUEZ

Q. Would the witness please state his name and business address?

A. Jared Rodriguez and my business address is 40 West 57th Street, Floor 23, New York, New York 10019.

Q. Mr. Rodriguez, by whom are you employed and in what capacity?

A. I am employed by Estates NY Real Estate Services LLC ("Estates"), which I will also sometimes refer to as "LeFrak," as Assistant Director, Utilities Conservation Engineering.

Q. How long have you been employed by LeFrak?

A. I have been employed by LeFrak for 7 years.

Q. Please briefly describe your educational and business experience.

A. I attended Tufts University and received a Bachelor of Science of Civil Engineering in 2008, and New York University, where I studied real estate development, sustainable development and finance, receiving a Master of Science degree in 2012. I founded an energy and sustainability consulting firm where I assisted clients with their plans for renewable energy planning, development and finance.

Q. Please generally describe your current responsibilities.

A. My responsibilities as Assistant Director, Utilities Conservation Engineering include identifying ways to foster energy efficiency throughout the residential and commercial real estate portfolio in existing buildings and in new construction, planning capital projects related to

energy-related buildings systems and developing a procurement strategy for utilities of all types.

Q. Of what local and regional energy bodies are you a member?

A. Member, Sustainability Committee, Real Estate Board of New York (REBNY); Member, 80x50 Buildings Partnership, Urban Green Council; Member, NYC 2030 District Multifamily Retrofit Working Group; Member, and the Carbon Neutral Buildings Roadmap Technical Advisory Group, NYSERDA.

Q. For which intervening party are you submitting testimony in Rate Case 16-G-0059?

A. I am testifying on behalf of Estates NY Real Estate Services LLC.

Q. Have you previously submitted testimony on behalf of Estates or any affiliate or subsidiary of Estates before the New York State Public Service Commission ("Commission")?

A. Yes, I testified on behalf of Estates in Case No. 16-G-0059.

Q. Please describe Estates and provide an overview of its properties.

A. Estates is the owner/manager of a very large real estate portfolio in the New York metropolitan region and in South Florida, Washington State and California. The firm owns and manages over 18,000 residential apartment units and over 20 million square feet of commercial office and retail space.

Q. Is Estates a customer of National Grid (The Brooklyn Union Gas Company dba National Grid NY) which is also referred to as the "Company" in this testimony?

A. Yes, the Estates properties include 38 buildings ("Buildings") which are, in the aggregate, one of the National Grid's largest gas customers taking Firm Service.

Q. What is the total estimated annual average gas usage of the Buildings under the Firm Service Classification?

A. 5,557,710 therms, which is the average of the most recent 3 years of annual usage.

PURPOSE OF TESTIMONY

Q. What is the purpose of your testimony in this National Grid rate case?

A. The purpose of my testimony is as follows:

1. To demonstrate the value of natural gas demand response to the Company, the Company's ratepayers and to all the citizens of New York State;

2. To lend support for the furtherance and expansion of the geothermal Non-Pipe Alternative ("NPA");

3. To support efforts to decarbonize the New York City and State energy system via electrification of building equipment which contributes to peak gas demand and capacity constraints;

4. To lend support for the furtherance of the efforts by National Grid to introduce and develop more sources of biomethane and renewable natural Gas ("RNG") and Hydrogen Gas via a Green Gas Tariff and other efforts; and

5. To make recommendations regarding the SC-3 rate class which serves many low income and family customers in New York City.

DISCUSSION

Q. Please state give a general overview of recent energy/environmental legislation and other agency efforts in New York State.

A. Recent legislation includes the Climate Leadership and Community Protection Act, NYISO carbon pricing, NYC Local Law 97, and additional efforts to modify NYSERDA or utility-managed incentive programs to further buildings energy decarbonization and efficiency.

Q. How do you think these initiatives will affect Estates moving forward?

A. My expectation is that these initiatives will require large capital (or other) expenditures at properties to modify or replace building equipment to comply with regulations. Owners of properties from smaller multi-family residential to large commercial have substantial sunk costs related to natural gas infrastructure and heating plant equipment; which burns

natural gas as a primary fuel. It is important to consider and calculate the full lifecycle costs that Customers will face as decarbonization efforts progress. In addition, customers unable to comply with NYC Local Law 97 because of the age of the building or other factors may be burdened with additional costs such as purchasing renewable energy credits (RECs) so as to achieve deductions from the required annual building emissions.

Q. Are there any solutions which would continue to use natural gas or biomethane?

A. Yes. Decarbonizing the natural gas supply is a possible solution to avoiding the substantial waste associated with the abandonment of any portion of the natural gas infrastructure (which infrastructure has been increasingly adopted in recent years by Customers) which would be a desirable outcome of decarbonization efforts. Such solutions would include the use of renewables such as agricultural or diary industry-sourced gaseous fuels like biomethane or hydrogen as discussed below.

Q. Do you favor supplementing gas network decarbonization by heating electrification efforts?

A. Heating electrification should be given serious consideration. Electrification would assist in avoiding expanding the natural gas network with its capacity constraints. One favorable method of managing decarbonization by electrification is expanded use of electric domestic hot

water equipment. Specifically, natural gas demand for the sole purpose of producing domestic hot water leads to network capacity constraints during periods of cold weather. This is caused by the peaking nature of domestic hot water demand in the morning coinciding with likely high natural gas demand response periods.

Q. How would you respond to an argument that carbon-producing fossil fuels including natural gas are used to generate electricity which would produce additional pollution thereby reducing or neutralizing decarbonization via conversation of gas-fired hot water equipment to electric power? As a corollary, how would you respond to an argument that such electrification may well necessitate the construction of additional generation plants and an expansion of the electric gird to accommodate the additional usage and demand that would follow electrification?

A. Taking these arguments one at a time, atmospheric pollution by carbon-producing combustion is more easily addressed in a baseline electric power plant than at individual gas burner tips. Also, baseload electric generation plants in NYS include sources that do not use fossil fuel such as hydro, nuclear and utility scale wind and solar. As to the possible necessity for the addition of electric utility generation, transmission and distribution plant, I note that these would address peak usage and my intent in this testimony is that the combination of initiatives and technologies that I am recommending

in this testimony (when expanded to include electricity - such as electric demand response) would reduce peak usage of both gas and electricity such that, for this reason, any significant expansion of electric utility plant would be unnecessary.

Q. Do you see other alternatives to natural gas?

A. Yes, the conversion of equipment to utilize hydrogen gas, including blended forms, remains an alternative as I discuss later in my testimony.

Q. Is gas demand response an effective means of helping the State and City to reach its goals?

A. Yes, the Company's pilot gas demand response program should be significantly expanded to help address the capacity issues and enable the State, City and the Company to reach their respective goals for maximizing the use of the system and reducing the overall amount of fossil fuels consumed.

Q. How will these goals be achieved by the use of gas demand response?

A. Gas demand response allows the Company to maximize its gas network to bring additional commodity gas to its Customers, thereby offsetting carbon intensive liquid fossil fuels and reducing carbon emissions. A gas demand response program incentivizes Customers to limit usage at times of capacity constraints via direct payments to participating

Customers which incentivize them to limit gas use generally and especially during such periods. This, in effect, monetizes timely responses during system constraint periods. Customers may utilize electrification and gas-to-electric switching as a means of achieving demand response goals, which not only helps to alleviate capacity limitation issues but also reduces carbon emissions. Peak period gas consumption, most likely driven by domestic hot water demand, appears to me to be an appropriate target for electrification in the near term.

Q Are there approaches or technologies, in addition to electrification, that you believe would be helpful to achieving demand response goals?

A. Load shifting is another approach to reducing demand. An example of load shifting technology is being implemented by companies such as RadiatorLabs, which modify radiator enclosures to store energy and release it on demand. It is possible to avoid natural gas consumption during peak gas periods with RadiatorLabs' technology. Pairing this technology with advanced electrification technologies can yield significant reductions in gas capacity constraints. RadiatorLabs' technology appears to be a "bridge" technology that is able to yield significant reductions in natural gas peak load now, while the building sector shifts to low-carbon heating technologies.

Q. Do you have recommendations to the Parties in this Proceeding regarding gas demand response?

A. We urge the Parties to consider methods of enhancing or incentivizing gas demand response via electrification as a means of achieving decarbonization goals, reducing overall costs to gas ratepayers, maximizing gas network efficiency and utilization, improving local air quality, and fostering a market for the development of new methods or technologies. We also urge the Company to expand its existing pilot gas demand response program and ultimately make the program permanent.

Q. How does the development and implementation of geothermal NPA heating help Estates and the overall energy marketplace in terms of energy efficiency, carbon reduction and resiliency?

A. Geothermal technology offers highly energy efficient production of domestic hot water and space heating. The technology, particularly shared loop geothermal energy systems, will likely play a key role in reinforcing the Company's ability to meeting decarbonization goals via heating electrification and reducing peak gas capacity. Without addressing potential legal steps necessary for the Company to engage a line of business producing and selling geothermal energy, the Company may be able to develop alternative revenue sources with an affiliated thermal company selling thermal energy as a means of offsetting possible losses incurred by

an overall reduction in fossil-based natural gas consumption. Alternatively, or concurrently, the production and sale of geothermal energy would also provide private businesses with opportunities resulting in job creation in New York State. Geothermal technology helps Estates NY meet regulatory requirements in that it provides a cost-effective low- to zero-carbon source of heat and remains the only likely technologically feasible alternative for the wholesale electrification of large multifamily and commercial buildings.

Q. Can you provide an explanation of a thermal company?

A. A thermal company is a company that builds, maintains and manages shared infrastructure (which would include locations in public rights of way and on private property) that provides a heating and cooling medium for water source heat pumps or similar technologies. A thermal company's distribution facility is most likely comprised of a horizontal water pipe network, heat exchangers, pumps and pumping stations, and physical connections to customers and heat sources or sinks. The thermal company, would provide a platform for sharing thermal energy across various customers, and might offer the possibility of establishing a bi-direction thermal distribution system similar to the bi-directionality of the electric distribution network envisioned by New York States Renewing the Energy Vision (REV) plan.

Q. What other benefits might a thermal company provide?

A. A thermal company may also provide a platform for new business ventures that provide heat sources or sinks via vertical bore fields (geothermal), thermal storage media, or via industrial waste. The New York City Department of Environmental Protection ("DEP") recently established an Office of Energy and Resource Recovery, which may facilitate recovery of energy resources via thermal energy extraction from the sanitary sewer network or production of Renewable Natural Gas from DEP-managed wastewater treatment facilities. If any applicable legal hurdles can be surmounted and this can be accomplished by the Company, such a model would be complementary to the Company's goals as expressed in the Future of Heat Panel's direct testimony, the Company's natural gas business, the State of New York's decarbonization goals and the goals of the Department of Public Service. I strongly encourage the Commission to look into establishing a regulatory framework for regulated monopolies to establish thermal company affiliates. Such a framework, either in conjunction with utility affiliates or independently, would provide new private business opportunities and hold the possibility of significantly reducing cost risks for Customers in light of the State's decarbonization goals.

Q. How would the development of more biomethane and RNG help the City and State reach their stated goals?

A. Renewable natural gas and hydrogen blending help New York City and State achieve decarbonization goals by moving natural gas toward a regime harmonized with the Renewing the Energy Vision (REV) framework. We strongly recommend that the Commission recognize the need for a gas REV framework to foster gas network bi-directionality, and an increase in local biomethane, RNG and hydrogen suppliers and sources, and to recognize the gas network's potential as a large-scale and seasonal energy storage and delivery system. Rethinking the gas network is vital to the State and City achieving its goals. Providing a pathway to, and a regulatory framework for, the Company forming a thermal company affiliate is desirable to in connection with natural gas being folded into REV.

Q. Please describe a typical SC-3 tenant and how increases in the SC-3 rate impact these families?

A. Estates NY's building stock is primarily comprised of rent stabilized apartments regulated by the New York State Department of Homes and Community Renewal "(DHCR"). Therefore, a high proportion of our tenants are low- and moderate-income residents. Higher natural gas and electric costs naturally burden each property, creating upward pressure on rents for those residents who do not reside in rent stabilized apartments. The Company's rate increases have typically outpaced rent increases,

reducing funds necessary to maintain these buildings and provide resident amenities.

Q Please discuss the current SC-3 rate and the potential problems with its level.

A. The Company is proposing an increase in base distribution rates of over 30% in the tail block for the SC-3 customer class. The overall increase in proposed bills to the typical SC-3 customer is pegged at 11%. (Exhibit RDP, Schedule 4 Page 8 of 18.). It is important for the Parties in this case to realize that this massive increase in base rates is hidden by low commodity costs which dampen the overall impact of the rate increase. There is no question that the natural gas market has been very favorable to the consumer over the past ten years. On the other hand, to expect that commodity costs will continue to stay low going forward and yet seek the proposed increases in customer bills is inconsistent. It is therefore important to call out the real increase here for the SC-3 customer will be in the range of 25% on a typical bill. This increase is on top of a greater than 25% increase in base rates agreed upon in the last rate case. This means that base rates will have increased by over 50% in less than four (4) years for this rate class, which primarily consists of low- and average-income family customers in New York City.

Q. What could happen to the overall bill impact if natural gas prices were to return to 10 year or even twenty-year averages?

A. If gas prices were to rise to even five-year averages, the average bill would then increase by over $1.00 per MMBtu as a result of the purchased gas passthrough mechanism which shifts all gas cost risk to the customer. Should this return to more historically normal gas prices occur, the overall bill impacts of this rate increase would be much larger.

Q. What is your recommendation to the Commission on the level of Rate SC-3?

A. The Commission should not be misled by the current environment of low gas prices in thinking that the rate increase sought by the Company is likely to be limited to only a 10% increase in rates. The Customers will bear ALL the risk associated with potentially higher gas commodity costs and, given the uncertainty of deliverability into New York during peak periods as well as the uncertainty associated with the price of the commodity at the wellhead, any projection of similar gas cost going forward is, at best, speculative. The Commission must understand that the increase in base rates for this class of customer is exorbitant unless the goal of the Commission and other New York State agencies is to raise the cost of natural gas delivery in order to shift customer behavior and enhance decarbonization goals. If this is an accurate assessment of the

future of heat and/or gas distribution in the state, the Commission must make provisions for Customers to feasibly and economically convert buildings to electrified heat or other low-carbon alternatives.

Q. What other concerns do you have related to Company's provision of natural gas delivery to your facilities?

A. If the goal of the Commission and New York State is to reduce the size of the Company's Customer pool, the Commission must recognize that, should this occur, it would necessarily increase the cost to remaining Customers via the Company's regulated recovery of fixed costs related to maintaining and/or expanding natural gas delivery capacity in the KEDNY service territory. If this conclusion is accurate, I urge the Commission to immediately cease approval of additional fixed cost acquisition by the Company and provide the Company access to additional Customers via new business models such as shared loop geothermal or thermal company ventures. Increasing or maintaining the size of the Customer pool would provide necessary price stability for those Customers that remain connected to the natural gas distribution network.

Q. Does that conclude your testimony?

A. Yes